034IN - FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL - A.Y. 2023-2024

Full Examination - July 19 2024

PART 1 - Problem 2

Tag List: `[STANDARD ASSESSMENT] [PARTIAL CREDIT] [SYMBOLIC MATH TOOLBOX] [CONTROL TOOLBOX]`

Consider the **linear system** described by the **transfer function**:

$$G(s) = \frac{80\,(s+1)}{(2-s)\,(s+2)}$$

The initial condition (i.e., the initial state) is the "*zero state*" $x(0) = \left[0\,,\,0\right]^{\top}$.

Assignment

Your code shall solve the following two problems regarding the dynamic system described by the above equations.

P1: The Theorem of the Initial Value for the Laplace Transform

- **Apply** to the system $G(s)$, with initial condition $x(0)$, the **input** $u(t) = 2\,t \cdot \cos(3\,t)\mathbf{1}(t)$, where $\mathbf{1}(t)$ is, as usual, the unit step function.
- **Calculate** the **Laplace** transform of the **output** $Y(s)$, and **save** the obtained expression in the **symbolic variable** **`Ys`**.
- **Applying the Initial Value Theorem** for the Laplace transform, **determine** the values $\mathbf{y(0)}$ and $\dot{\mathbf{y}}\mathbf{(0)}$ (i.e., the value of the output and the first derivative of the output at the time instant $t = 0$) from the expression of the transform $Y(s)$.
- Store the resulting values in the **symbolic variables** **`y0L`** and **`Dy0L`** respectively.
- From the Laplace transform $Y(s)$ **evaluate** the corresponding **time-function** $y(t)$. Assign the resulting expression to the symbolic variable **`y_t`**.
- Using **`y_t`**, **evaluate** the time first derivative $\dot{y}(t)$ of $y(t)$ at the time instants $t_i = \left\{0, \frac{1}{10}, \frac{1}{4}, \frac{1}{2}, \frac{\pi}{3}, \frac{2\pi}{3}, \pi\right\}$. Save the resulting values in the array **`D_y_t`**.

Solution

Solution to the Problem P1

Let's define the transfer function $G(s)$ and the input signal $u(t)$ as symbolic variables:

```
syms s t
```

```
u_t = 2*t*cos(3*t)*heaviside(t) % <<<<---- the given input signal u(t)
```

u_t = $2\,t\cos(3\,t)\,\mathrm{heaviside}(t)$

```
Us = laplace(u_t,s)    % <<<<---- the corresponding Laplace transform U(s)
```

Us =

$$\frac{4\,s^2}{(s^2+9)^2}-\frac{2}{s^2+9}$$

```
% --- the LTI system transfer function ---
Gs = 80*(s+1)/((2-s)*(s+2))
```

Gs =

$$-\frac{80\,s+80}{(s-2)\,(s+2)}$$

Now let's calculate the Laplace transform of the "forced" output:

```
Ys = Gs*Us
```

Ys =

$$\frac{\left(\frac{2}{s^2+9}-\frac{4\,s^2}{(s^2+9)^2}\right)(80\,s+80)}{(s-2)\,(s+2)}$$

The Initial Value Theorem

$$\lim_{t\to 0} y(t) = \lim_{s\to\infty} Y(s)$$

Refer to the Laplace transform related material of the course for details.

Thus

```
sYs = s*Ys
```

sYs =

$$\frac{s\left(\frac{2}{s^2+9}-\frac{4\,s^2}{(s^2+9)^2}\right)(80\,s+80)}{(s-2)\,(s+2)}$$

```
y0L = limit(sYs, s, Inf)
```

y0L = 0

The Laplace Transform of the First Time Derivative

Now we are able to compute the Laplace transform of the output first time derivative

```
D_ys = s*Ys - y0L % <<<--- the Laplace transform of dot{y}
```

D_ys =

$$\frac{s\left(\frac{2}{s^2+9}-\frac{4\,s^2}{{\left(s^2+9\right)}^2}\right)\,\left(80\,s+80\right)}{\left(s-2\right)\,\left(s+2\right)}$$

```
sDys = s*D_ys
```

sDys =

$$\frac{s^2\left(\frac{2}{s^2+9}-\frac{4\,s^2}{{\left(s^2+9\right)}^2}\right)\,\left(80\,s+80\right)}{\left(s-2\right)\,\left(s+2\right)}$$

```
Dy0L = limit(sDys, s, Inf)
```

Dy0L = 0

The Function in the Time Domain

From $Y(s)$ to $y(t)$

```
y_t = simplify(ilaplace(Ys,s,t))
```

y_t =

$$\frac{200\,e^{-2\,t}}{169}-\frac{800\cos(3\,t)}{169}+\frac{600\,e^{2\,t}}{169}-\frac{960\sin(3\,t)}{169}+\frac{160\,t\cos(3\,t)}{13}-\frac{480\,t\sin(3\,t)}{13}$$

```
Dyt = diff(y_t, t)
```

Dyt =

$$\frac{1200\,e^{2\,t}}{169}-\frac{400\,e^{-2\,t}}{169}-\frac{800\cos(3\,t)}{169}-\frac{3840\sin(3\,t)}{169}-\frac{1440\,t\cos(3\,t)}{13}-\frac{480\,t\sin(3\,t)}{13}$$

```
tVALS = sym([0, 1/10, 1/4, 1/2, pi/3, 2*pi/3, pi]);

D_yt_array = subs(Dyt, t, tVALS);
disp(D_yt_array')
```

$$\begin{pmatrix} 0 \\ \frac{1200\,e^{1/5}}{169} - \frac{400\,e^{-\frac{1}{5}}}{169} - \frac{2672\cos\left(\frac{3}{10}\right)}{169} - \frac{4464\sin\left(\frac{3}{10}\right)}{169} \\ \frac{1200\,\sqrt{e}}{169} - \frac{400\,e^{-\frac{1}{2}}}{169} - \frac{5480\cos\left(\frac{3}{4}\right)}{169} - \frac{5400\sin\left(\frac{3}{4}\right)}{169} \\ \frac{1200\,e}{169} - \frac{400\,e^{-1}}{169} - \frac{10160\cos\left(\frac{3}{2}\right)}{169} - \frac{6960\sin\left(\frac{3}{2}\right)}{169} \\ \frac{480\,\pi}{13} - \frac{400\,e^{-\frac{2\pi}{3}}}{169} + \frac{1200\,e^{\frac{2\pi}{3}}}{169} + \frac{800}{169} \\ \frac{1200\,e^{\frac{4\pi}{3}}}{169} - \frac{400\,e^{-\frac{4\pi}{3}}}{169} - \frac{960\,\pi}{13} - \frac{800}{169} \\ \frac{1440\,\pi}{13} - \frac{400\,e^{-2\pi}}{169} + \frac{1200\,e^{2\pi}}{169} + \frac{800}{169} \end{pmatrix}$$